**Course Code:** COP4803

**Course Name**: Keyvan AVIATION Navigating Future Skies

# SYLLABUS[1]

**Instructor**

| | |
|---|---|
| Office | : North Campus A Block 4th Floor |
| E-Mail | : tuzuntolga.inan@bau.edu.tr |
| Office Hours: | : Weekdays |
| CV (link) | : https://cdn.bau.edu.tr/staff/2125.pdf |

**Course Information**

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|---|---|
| Period | : Spring |
| Time | : Tuesday / 11:30-14:20 |
| Course Credit / ECTS | : 3/6 |
| Classroom | : KA201 |
| Mode of Delivery: | : Face to Face |
| Course type | : Departmental Elective |
| Course ECTS Page Link | : |

**Prerequisite (If Available)**

There is no preliminary preparation.

**Course Objectives**

This training is set at an advanced level and is intended to cover all aspects of aviation security from a management and senior management perspective. It will include actual topics such as terrorism, sabotage, direct and indirect threats, modern slavery/human trafficking, cyber security and information security.

**Course Learning Outcomes**

The students who succeeded in this course, will learn the terms below. These are classified as;

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| Understand the fundamental principles and applications of Aeronautical Navigation Data. |
| Analyze the processes involved in the generation, update, and management of navigation data. |

[1] 1 It is essential that the syllabus announced at the beginning of the term is not changed except when necessary. When a requirement occurs, the syllabus can be changed by the instructor of the course by notifying this situation in writing or verbally beforehand. It is students' responsibility to follow the current syllabus.

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| Learn the insights of the various technological and operational aspects in Sustainable Aviation. |
| Explain the role and impact of projects like SESAR in advancing sustainable aviation practices. |
| Develop the ability to critically assess and propose improvements in aviation data systems and sustainability asures. |
| Acquire the practical skills in applying theoretical knowledge to real-world scenarios in aviation technology sustainability. |

**Contribution of the Course to the Program**

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**Course Structure**

Course will be given face to face.

**Course Policies**

***Communication Channels and Methods:*** All the students can reach on itslearning and mail.

***Usage of Digital Tools:*** Mobile technologies such as mobile phones, tablet computers, laptop computers can only be used for teaching purposes. Please respect the lecturer and your friends by turning off the volume or turning off your mobile phone. Your cell phones should be muted or turned off before class. There may be some important situations in our lives (health, family, personal) in such cases, please turn your phone to vibration and if you really need to talk, leave the class and talk outside and come back to the lesson

***Assignents and Project Deadline:*** All homework will be sent via Itslearning (you can select Bahcesehir Ugur Education Institutions from the list on www.itslearning.com , enter your username and password). All homework will be sent via Itslearning (you can select Bahcesehir Ugur Education Institutions from the list on www.itslearning.com , enter your username and password). Sufficient time has been calculated and given to you to complete your homework. If you are unable to do your homework, please contact the instructor of the course immediately and report your situation. You must submit your homework on time. Your assignments are due one day before the lesson. For each late submission, the grade will be reduced by 10% (per day). I wish you all success in a new era!

***Attendance:*** Students are expected to attend scheduled classroom sessions and participate in a positive and meaningful way. Getting the most out of each classroom session requires you to read the provided materials beforehand. If participation is not possible or expected, please provide this information before your absence. If it is

not possible to attend the lesson and it occurs unexpectedly on the day of the lesson, please inform your instructor by e-mail.

***Disabled Student Support:*** You can contact me directly regarding the issues that may be an obstacle for you (vision, hearing, etc.). In addition to this, there is a Disabled Student Unit in order to minimize the difficulties that our disabled students will encounter due to their disabilities and to eliminate the obstacles. You should contact this unit regarding your situation. Click to access the web page of this unit.

***Oral and Written Communication Ethics:*** During the lesson you have to express yourself respectfully in your communication with your friends and with me. In addition, you are responsible for maintaining this respect in discussions, homework and correspondence on the online platform

***Privacy and Copyright:*** In accordance with the Personal Data Protection Law, the courses will be recorded on the online platform within the scope of your approval and knowledge. In addition, it is strictly forbidden to register the participants (students and instructors) during the course.

**Course Resources**


79th Annual General Meeting and World Air Transport Summit . (2023 ). IATA Annual Review 2023 . IATA .

Annex 15. (n.d.). In ICAO, The Convention on International Civil Aviation Annexes.

ARINC 424-22, Navigation System Database. (n.d.).

Aviation Intelligence Unit . (2021 ). Flying the 'perfect green flight': How can we make every journey as environmentally friendly as possible? EUROCONTROL Think Paper #10.

Convention on International Civil Aviation 9th Edition. (2006 ).

Convention on International Civil Aviation Original Document . (1944).

DO-200B, Standard For Processing Aeronautical data. (n.d.).

DO-201A, User Requirements for Navigation Data. (n.d.).

DO-201B, User Requirements for Navigation Data. (n.d.).

DO-272D, User Requirements for Aerodrome Mapping Information. (n.d.).

(07.09.2023). EBBR AIP. Belgian Civil Aviation Authority.

ED-76A, Standard For Processing Aeronautical data. (n.d.).

EUROCONTROL. (2022). Aviation Outlook 2050 . EUROCONTROL.

EUROCONTROL. (2023). Aviation Sustainability Briefing . EUROCONTROL .

Eurocontrol Emission Calculator 2021. (n.d.). Retrieved from Eurocontrol: https://www.eurocontrol.int/sites/default/files/2021-12/eurocontrol-emission-calculator-2021-update.xlsx

(2023). EUROPEAN AVIATION OVERVIEW . EUROCONTROL.

IATA Carbon Offset Program. (n.d.). Retrieved from IATA: https://www.iata.org/en/programs/environment/carbon-offset/

ICAO. (2018). 10066 Aeronautical Information Management. ICAO.

ICAO. (n.d.). Annex 5 - Units of Measurement to be Used in Air and Ground Operations.

ICAO Doc 9674 – WGS 1984 Manual. (n.d.).

ICAO Doc. 8126 AIS Manual. (n.d.).

ICAO-UNDP-GEF. (2018). Sustainable Aviation Fuels Guide .

REGULATION (EU) 2017/373 . (2017). Official Journal of the European Union .

SESAR Joint Undertaking . (2022). SESAR 3 Joint Undertaking Multiannual Work Programme . Luxembourg : Publications Office of the European Union.

Turkish Civil Aviation Act. (1983).

## Grading and Evaluation

| Assignment | Description | Scoring | Weight (% |
|---|---|---|---|
| * Midterm | It covers the topics we covered until the relevant week. | 100 | 40 |
| *Presentation | Presentation | 100 | 10 |
| * Final Exam | It covers the topics we covered until the relevant week. | 100 | 50 |
| TOTAL | | 100 | 100 |

| Semester Requirements | Number of Activities | Level Of Contribution |
|---|---|---|
| Presentation | 1 | 10 |
| Midterm | 1 | 40 |
| Final | 1 | 50 |

## Course Calendar

| Week/Place | Course Topic | To Do | Assignments Deadline* |
|---|---|---|---|
| **Week** | Subject | Related Preparation | |
| **1** | Introduction, review of the syllabus,plan of the course and Introduction to Company and Industry Overview<br>Introduction to International Regulations on Civil Aviation | Presentation | |
| **2** | **Introduction to Aeronautical Navigation Data**<br>Fundamentals of aeronautical navigation data: Definition and importance.<br>Historical development and evolution of navigation systems in aviation.<br>Types of navigation data: waypoints, navaids, airspaces, and procedures. | Presentation | |
| **3** | **Data Sources, Processing, and Standards**<br>Overview of data sources for aeronautical navigation (e.g., AIP, NOTAMs).<br>Data processing techniques: From raw data to usable information.<br>International standards and regulations governing navigation data (e.g., ICAO standards). | Presentation | |
| **4** | **Navigation Database Systems and Applications**<br>Structure and functioning of aeronautical navigation databases.<br>Integration of navigation data into avionic systems (e.g., FMS, ATC systems).<br>Real-world applications: Flight planning, air traffic management, and simulation. | Presentation | |
| **5** | **Challenges and Future Trends in Aeronautical Navigation Data**<br>Current challenges in navigation data management (e.g., data accuracy, real-time updates).<br>The impact of emerging technologies on navigation data (e.g., ADS-B, GNSS).<br>Future directions in navigation data: Innovations and potential developments. | Presentation | |

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| **6** | **Airspace Management**<br>Understanding different classes of airspace and their regulations.<br>Techniques for managing air traffic flow efficiently.<br>Coordination between air traffic control and flight operations.<br>Use of technology in monitoring and navigating airspace. | Presentation |
| **7** | **Ethics and Responsibility in Aviation**<br>Key ethical principles in aviation: safety, integrity, responsibility.<br>Case studies on ethical decision-making in aviation scenarios.<br>Professional responsibility and ethical standards in aviation operations.<br>Global and cultural aspects of aviation ethics.<br>Environmental ethics and sustainability in aviation. | Presentation |
| **8** | **Modern Airspace Integration Between Civil Aircraft, Military Aircraft and UAVs**<br>Analyzing airspace use by civil, military, and unmanned aerial vehicles (UAVs).<br>Coordination methods for shared airspace and conflict avoidance.<br>Technological systems supporting integrated airspace management.<br>Regulatory frameworks governing multi-use airspace operations. | Presentation |
| **9** | **MIDTERM** | |
| **10** | **Keyvan Aviation HQ Exploration Day: HQ Tour** | Workshop |
| **11** | **Fundamentals of Sustainable Aviation**<br>Introduction to the concept of sustainable aviation.<br>The historical evolution and significance of sustainability in aviation.<br>Economic and environmental motivations behind sustainable practices. | Presentation |
| **12** | **Regulations and Global Initiatives in Sustainable Aviation**<br>Overview of international and national regulations for sustainable aviation.<br>Role of aviation authorities and government policies.<br>Discussion on major global initiatives and agreements (e.g., ICAO's CORSIA). | Presentation |

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| **13** | **Technological Innovations and Future of Sustainable Aviation**<br>Exploration of technological advancements (e.g., electric aircraft, alternative fuels).<br>In-depth study of significant sustainable aviation projects (SESAR, Clean Sky).<br>Future trends, potential challenges, and the road ahead for sustainable aviation. | Presentation |
| **14** | **Presentation and Assesment of the Final Projects** | Presentation |
| **15** | **Final Exam** | |

**Matters Needing Attention**

Students are required to attend the classes on time, to avoid disturbing behaviors and to comply with the issues that they have to attend the midterm exam, homework, case studies and final exam. It is important to get permission before taking the floor in lessons that will be taught interactively and not to disturb the concentration of other students and lecturers.

In addition, the students who will take the course;

- Not being late for class,
- Active participation in classes with a minimum 70% attendance rate,
- Not making any physical, psychological or verbal violence or slang expressions,
- To comply with disciplinary penalties,
- In case of emergency, they are required to inform the faculty member about the subject beforehand.

**Academic Integrity, Cheating and Plagiarism**

Hexham (2005) defines plagiarism as a planned deliberate action to deceive the reader by pretending to be someone's word or words. Academic plagiarism is also in the form of an author's use of more than four words in his own research, without using quotation marks from a written source, without a precise reference to the original source published before, or in the form of similar fraudulent behavior in scholarship-application forms he wrote to contribute to his research.

• Actions that can be defined as contrary to publication ethics in the scientific community are defined as follows;

- plagiarism, - cheating, - paraphrasing,
- fabrication and falsification of data,
- to help copying and plagiarism, -To prevent others from accessing a source or data,
- Appearing as a writer in joint studies without contributing,
- Use of widely known / anonymous information, -No regular attribution,

- Self-plagiarism etc.

• Not everything on the Internet is public and cannot be obtained without permission or reference.

• Studies conducted without proper reference are graded with a score of zero.

• Large amounts of manuscripts copied without being quoted will be considered plagiarism and you will be responsible.

• Please be aware that the penalties for plagiarism can range from grading homework to dropping you out of class.

• If you copied, plagiarized or copied / pasted, do not expect the instructor of the course to write you a reference letter or to be your advisor.

***• How is plagiarism penalized?***

If it is revealed that you have been part of any misconduct during the course, the instructor of the course will inform the program coordinator. Depending on the seriousness of the situation, the Program Coordinator decides with the committee the appropriate penalty from giving a grade of 0 for the assignments to discharging you from the course. In any case, the student has the right to defend him/herself.

***ARTICLE 25 – (1)*** *In case it is doubled that a student cheats or attempts to cheat, commits plagiarism or similar violations defined in the applicable disciplinary regulation in any exam, assignment or other assessment activities, a disciplinary proceeding is brought against the student. Such activity is not assessed during the proceedings. A student who is found guilty is assigned zero point in addition to the disciplinary punishment. If the student is found innocent as a result of disciplinary proceeding, the exam taken by the student shall be assessment or a make-up exam or activity is provided.*

*You can access Bahçeşehir University and Higher Education Institution Regulations by clicking this sentence.*

**Prepared by Name Surname and Date of Preparation:**

Asst. Prof. Dr. Tüzün Tolga İnan

09.02.2026